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Biochemistry

Msn2p, a zinc finger DNA-binding protein, is the transcriptional activator of the multistress response in *Saccharomyces cerevisiae*

(stress response element/heat shock/DNA damage)

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ABSTRACT **The stress response promoter element (STRE) confers increased transcription to a set of genes following environmental or metabolic stress in *Saccharomyces cerevisiae*. A λ gt11 library was screened to isolate clones encoding STRE-binding proteins, and one such gene was identified as *MSN2*, which encoded a zinc-finger transcriptional activator. Disruption of the *MSN2* gene abolished an STRE-binding activity in crude extracts as judged by both gel mobility-shift and Southwestern blot experiments, and overexpression of *MSN2* intensified this binding activity. Northern blot analysis demonstrated that for the known or suspected STRE-regulated genes *DDR2, CTT1, HSP12*, and *TPS2*, transcript induction was impaired following heat shock or DNA damage treatment in the *msn2*-disrupted strain and was constitutively activated in a strain overexpressing *MSN2*. Furthermore, heat shock induction of a STRE-driven reporter gene was reduced more than 6-fold in the *msn2* strain relative to wild-type cells. Taken together, these data indicate that Msn2p is the transcription factor that activates STRE-regulated genes in response to stress. Whereas nearly 85% of STRE-mediated heat shock induction was *MSN2* dependent, there was significant *MSN2*-independent expression. We present evidence that the *MSN2* homolog, *MSN4*, can partially replace *MSN2* for transcriptional activation following stress. Moreover, our data provides evidence for the involvement of additional transcription factors in the yeast multistress response.**

Stress responses are ubiquitous among living cells. Chemical agents, temperature, osmotic shock, and nutrient depletion are among a highly diverse group of environmental conditions that alter patterns of gene expression in prokaryotic and eukaryotic cells. Among the most thoroughly studied of these stress responses is the transcriptional activation of heat shock genes following a transient increase in growth temperature. Recently, we identified a heat shock pathway in *Saccharomyces cerevisiae* that activates gene expression independently of the heat shock transcription factor or its promoter element (1, 2). This stress response differs from the classical heat shock regulatory response in that gene transcription is efficiently stimulated by a remarkable variety of stresses that includes in addition to heat shock, DNA alkylation, osmotic shock, oxidative damage, heavy metal exposure, and certain types of nutrient limitations (ref. 3; J. M. Treger, T. Magee, and K.M., unpublished data). Promoter disection of one yeast gene controlled by this multistress pathway, the DNA damage responsive gene *DDR2*, identified the nonpalindromic pentanucleotide CCCCT (C_4T) as the upstream element responsible for stress activation (2). A *DDR2*-derived C_4T-containing sequence (oligo 31/32) was shown to confer heat shock inducibility to a reporter gene (1) and mutations within the C_4T elements abolished reporter induction (2). The C_4T promoter element [also called the stress response element (STRE)] has been implicated in the multistress regulation of several yeast genes in addition to *DDR2*. These include the DNA damage-responsive gene *DDR48* (J. M. Treger and K.M., unpublished data), the yeast polyubiquitin gene *UBI4* (J. Simon and K.M., unpublished data), the cytoplasmic catalase T gene *CTT1* (3), the *TPS2* gene encoding the enzyme trehalose phosphate phosphatase (4), and *HSP12* (5).

Yeast extracts contain a 97-kDa protein (revised from 140 kDa reported previously) that specifically recognizes the C_4T motif and was proposed to be the transcription factor responsible for STRE-regulated expression of *DDR2* and other stress-responsive genes (ref. 2; unpublished data). Consistent with this idea were results obtained using both a C_4T-containing DNA sequence and a mutant A_4G-containing sequence that showed a correlation between disruption of STRE binding *in vitro* and abolition of transcriptional activation *in vivo* (2). The results presented here show that the major STRE-specific binding factor from yeast is the product of the *MSN2* gene, which was originally isolated as a multicopy suppressor of a *snf1-ts* mutation, and is predicted to be a C_2H_2 zinc-finger DNA-binding protein (6). Deletion of the *MSN2* gene greatly decreased the level of stress induction for all STRE-regulated genes tested, and overproduction of Msn2p led to constitutive activation of these same target genes. While *MSN2* encodes the primary STRE regulatory protein in yeast, our results indicate that the *MSN2* structural homolog *MSN4* can partially suppress a *msn2* mutation when it is expressed from a high copy number vector. An additional factor has also been identified that is distinct from Msn2p and Msn4p but binds specifically to C_4T elements and is predicted to contain C_2H_2 zinc-finger DNA-binding motifs.

MATERIALS AND METHODS

Yeast Strains and Media. Strain S288C (*MATα SUC2 mal mel gal2 CUP1*) was used as the wild-type control for DNA-binding experiments and RNA analysis. The *msn2* disruption strain MCY2144 (*MATα ade2-107 ura3-52 lys2-801 msn2Δ3::HIS3 his3Δ200*) was obtained from Marian Carlson (Columbia University, New York). This strain was transformed (7) with plasmids pEY32H and pEL45 (6) and URA^+ cells selected to generate the *MSN2* overexpressing strain MCY2144/pEY32H and the *MSN4* overexpressing strain MCY2144/pEL45, respectively. Strains for β-galactosidase assays were generated by

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Abbreviation: STRE, stress response element.
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transforming plasmids pCT and pCT oligo 31/32 (1) into wild-type strain M12B (*MATα trp1-289 ura3-52 gal2*) and strain MCY2144. Transformants were grown in synthetic complete medium lacking uracil (8). All other strains were grown in 1% Bacto-yeast extract/2% Bacto-peptone/2% dextrose.

Screening of Yeast Genomic DNA Library. A yeast genomic library in λ gt11 (Clontech) was screened according to manufacturer's instructions. After transfer of plaque proteins to nitrocellulose, filters were rinsed with binding buffer (100 mM KCl/25 mM Hepes, pH 8.0/1 mM dithiothreitol/0.05% Nonidet P-40/5% glycerol), then denatured in binding buffer containing 6 M guanidine hydrochloride and renatured as described (2). Concatameric probe (see below) was added to binding buffer containing 2.5% dry milk at a final specific activity of 1×10^5 cpm per ml. Filters were incubated with probe for 2 h at 4°C, then washed and exposed to film as described (2).

Concatamer DNA Synthesis. Concatamers were generated using a modification of the thermocycler-based method described by Hemat and McEntee (9). Two complementary dimeric primers (10 ng) were used as starting material. Primers for wild-type concatamers were: wcp1, 5′-G<u>CCCCT</u>GAATTCAGAGTTAGG<u>CCCCT</u>GAATTCAGAG-3′; wcp2, 5′-CTCTGAATTC<u>AGGGG</u>CCTAACTCTGAATTC<u>AGGGG</u>C-3′. Mutant concatamer primers used to analyze phage plaques were identical except that the C_4T sequences (or their complements AG_4) were changed to A_4G (or CT_4). Mutant concatamers used for Southwestern blot analysis contained C_4A (or TG_4) in place of C_4T. The reactions (100 μl) also contained 1 mM $MgSO_4$, 0.5 mM each of dCTP, dGTP, and TTP, 0.12 mM dATP, 150 μCi [α-^{32}P]dATP (3000 Ci/mmol; 1 Ci = 37 GBq), 6 units Vent polymerase (New England Biolabs), and 1× Vent polymerase buffer supplied by the manufacturer. Reactions were heated to 95°C for 10 min before the addition of polymerase, then cycled 30 times (65°C, 4 min; 95°C, 1 min) and terminated (65°C, 5 min). Product DNA ranged from 1 kb to greater than 10 kb with a yield of ≈20 μg per reaction. Concatamers were separated from unincorporated nucleotides using NICK columns (Pharmacia).

Cloning and Sequencing. λ DNA was isolated from a homogeneous stock of clone S2b using the protocol provided by Clontech. A *Bsi*WI restriction fragment containing the entire yeast genomic insert was subcloned into pBluescript SK$^+$ and sequenced by the dideoxy chain termination method with [α-^{35}S]dATP and Sequenase (United States Biochemical). Sequence alignment was performed using the BLAST e-mail server (10).

Cell Extract Preparation. Cells were grown at 30°C to OD_{595} 2–4 and harvested. Pellets were suspended in buffer A [50 mM Hepes, pH 8.0/0.4 M $(NH_4)_2SO_4$/1 mM EDTA/5% glycerol], disrupted mechanically with glass beads, and clarified (11). Extracts used for mobility-shift assays were immediately dialyzed against buffer B (75 mM KCl/25 mM Hepes, pH 8.0/1 mM dithiothreitol/1 mM EDTA/12% glycerol). Extracts used for Southwestern blot assays were concentrated by precipitation with 50% $(NH_4)_2SO_4$ saturation before dialysis against buffer B.

DNA-Binding Assays. Mobility-shift binding reactions (40 μl) were performed in the binding buffer used for library screening but containing 75 mM KCl. Yeast extract (60 μg) was incubated with 0.25 ng (30,000 cpm) of DNA probe and separated on a native 4% polyacrylamide gel as described (1). End-filled double-stranded oligonucleotide B (35 bp) was used as probe (2). Mutant competitor oligonucleotide B-C_4A is identical to B except that the single C_4T binding site has been changed to C_4A. Competitor DNA was added at 200-fold molar excess. For Southwestern blot analysis, cell extracts (60 μg) were electrophoresed on an 8% SDS polyacrylamide gel and electrotransferred to nitrocellulose membrane. Proteins were denatured, renatured, and probed as for library screening, except that the binding buffer contained 0.6% Nonidet P-40.

RNA Preparation and Northern Blot Analysis. For heat shock, cells were grown at 23°C to OD_{595} 0.2–0.3 and shifted to 37°C for 20 min. For DNA damage, cells were grown to OD_{595} 0.2–0.3 at 30°C and exposed for 60 min to 0.1% methyl methanesulfonate (Aldrich). RNA was extracted after glass bead-disruption as described (12). Denatured RNA (25 μg) was electrophoresed, blotted to nylon membrane, and hybridized to multiprimed ^{32}P-labeled probes using standard methods. *DDR2* transcript was probed with the 1.45-kb *Hin*dIII fragment from plasmid pBRA2 (12). *RNR3* transcript was probed with the 1.7-kb *Eco*RI/*Pvu*II fragment from plasmid pSZ214 (13). The remaining probes were PCR-amplified from selected regions of appropriate genes using M12B colonies as templates. The regions amplified were: nucleotides +57 to +556 of *CTT1*, +1 to +321 of *HSP12*, +1 to +551 of *TPS2*, +1558 to +1930 of *SSA3*, and +1 to +714 of *ACT1*.

Assay of β-Galactosidase. Cells were grown at 23°C to OD_{595} 0.2–0.3 and then heat-shocked for 1 h at 37°C and harvested. β-galactosidase activity (nmoles of *o*-nitrophenyl β-D-galactoside hydrolyzed per min per mg of protein) was assayed from crude extracts as described (1).

RESULTS

Isolation of Multiple C_4T-Binding Activities by Expression Library Screening. A λ gt11 expression library containing *S. cerevisiae* genomic inserts was screened in duplicate with ^{32}P-labeled C_4T-containing DNA concatamers and a total of 18 independent positive plaques were isolated from the equivalent of 4.5 genomes of correctly-oriented, in-frame fusions to *lacZ*. Six of these plaques were selected and purified through a second round of enrichment using both the C_4T-containing concatameric probe and a mutant probe in which the C_4T pentanucleotide was changed to AAAAG (A_4G). All six plaques showed strong binding to the C_4T concatamers but no detectable binding to the A_4G-containing probe (Fig. 1).

Phages S2b and S1a were purified to homogeneity and used to prepare λ lysogens in *Escherichia coli* strain Y1089 (14). Both S2b and S1a lysogens expressed STRE-binding activities as judged by Southwestern blot assays using C_4T concatamers as probe. Additionally, gel mobility-shift assays using radiolabeled oligonucleotide B (2) detected a STRE-binding activity in extracts prepared from lysogen S2b that was absent from nonlysogenic cells (data not shown).

Of the 18 candidate phage isolated in the screen, DNA sequencing and hybridization showed that 14 contained fragments related to phage S2b, whereas the remaining four were homologous to the insert in phage S1a (A.P.S., J. M. Treger, and K.M., data not shown). Expression of the STRE-binding activity encoded by phage S2b in *E. coli* was independent of insert orientation, indicating that transcription of this region likely initiated from a bacterial promoter internal to the yeast DNA fragment.

The *MSN2* Gene Encodes a C_4T-Binding Activity. A partial sequence of the 1.3-kb yeast DNA fragment in phage S2b showed complete identity to a region at the *S. cerevisiae MSN2* locus that overlapped the C-terminal 1200 nucleotides (roughly 60%) of the *MSN2* coding region and 140 nucleotides of its 3′ flanking region. The *MSN2* gene encodes a zinc-finger DNA-binding protein and was isolated by Estruch and Carlson (6) as a multicopy suppressor of a *snf1* kinase deficiency. The *SNF1* kinase is required for growth on certain nonglucose carbon sources (15). Disruption of the *MSN2* gene in the presence of functional *SNF1* did not affect cell growth on nonglucose carbon sources (6).

The DNA sequence of the insert in phage S1a was identical to an open reading frame encoding another putative zinc-

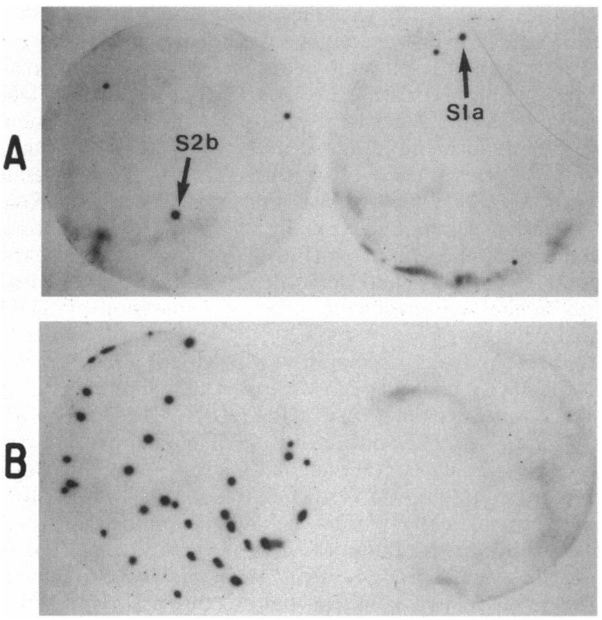

FIG. 1. Expression library screening identifies λ gt11 clones encoding STRE-binding activities. (*A*) Phage from a *S. cerevisiae* genomic library were plated at 30,000 pfu per plate and screened using a ^{32}P-labeled concatameric C_4T probe. The autoradiograph of two filters is shown. The arrows identify the positions of clones S2b and S1a. (*B*) Phage S2b was plated at 5,000 pfu per plate and purified through a second round of screening. Duplicate filters were probed with C_4T concatamers (*Left*) or mutant A_4G concatamers (*Right*).

finger DNA-binding protein of unknown function in yeast (unpublished data).

The Level of STRE-Binding Activity *in Vitro* Correlates with the Level of *MSN2* Expression in *S. cerevisiae*. As shown in lane 1 of Fig. 2, STRE-binding activity was readily detected using extracts prepared from $MSN2^+$ cells and radiolabeled B fragment in a gel mobility-shift assay. Binding was efficiently competed by an excess of unlabeled B fragment but was poorly competed by mutant competitor B-C_4A, in which the C_4T element was replaced with C_4A. The C_4A pentanucleotide does not confer stress inducibility to a reporter gene (J. M. Treger and K.M., unpublished data). The protein–DNA complexes detected appeared to consist of at least two components: a more intense and discrete slower mobility band and a less intense and more diffuse faster migrating species, both of which were efficiently competed by wild-type but not mutant oligonucleotides. Extracts prepared from MCY2144 cells containing the *msn2* gene disruption lacked the more intense slower mobility complex formed with labeled B fragment, but retained the diffuse, faster migrating species (lane 2). Furthermore, when extracts were prepared from MCY2144/pEY32H cells containing the *MSN2* gene in multiple copy, there was a significant increase in the intensity of the slower mobility complex (lane 3). These results argue strongly that the major C_4T-binding activity in yeast crude extracts was Msn2p.

Further mobility-shift binding reactions were performed using extracts prepared from strain MCY2144/pEL45, which contains the *msn2* disruption as well as *MSN4* on a high copy vector. *MSN4* is a structural homolog of *MSN2*, and its product is predicted to contain two tandem C_2H_2 zinc fingers near its C terminus (6). Overexpression of *MSN4* in this strain led to significantly more C_4T-binding activity than was present in the *msn2* disrupted strain alone (Fig. 2, lane 4). The binding activity produced by *MSN4* overexpression comigrated with the faster migrating species that was independent of *MSN2* expression, indicating that this second binding activity might be Msn4p. However, disruption of *MSN4* did not eliminate this species (data not shown), suggesting that factors distinct from Msn4p contributed to this STRE-binding activity.

Additional evidence linking the major C_4T-binding activity from yeast to *MSN2* was obtained by Southwestern blot analysis. Four proteins in $MSN2^+$ cell extract were detected that bound to C_4T concatamers (Fig. 3, lane 1). Of these, only one showed significantly better binding to the C_4T concatamers than to mutant C_4A concatamers (lane 4). This C_4T-specific binding

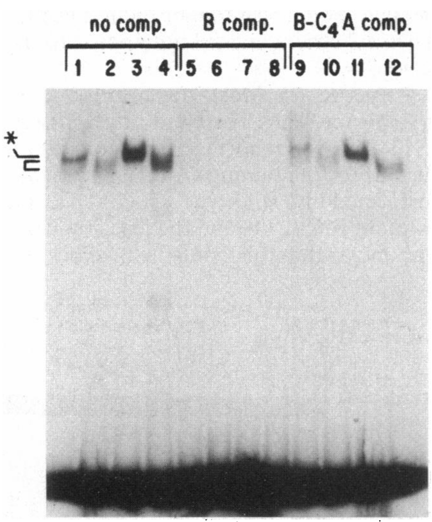

FIG. 2. The levels of STRE-binding activity correlate with expression levels of *MSN2*. Yeast extract (60 μg) was incubated with ^{32}P-labeled B fragment and electrophoresed. Lanes 1, 5, and 9, strain S288C ($MSN2^+$); lanes 2, 6, and 10, strain MCY2144 (*msn2*); lanes 3, 7, and 11, strain MCY2144/pEY32H (*MSN2* overexpression); lanes 4, 8, and 12, strain MCY2144/pEL45 (*MSN4* overexpression). The position of the Msn2p specific complex is indicated by an asterisk, and the diffuse, faster migrating complex(es) are shown with a bracket. Unbound DNA is visible at the bottom of the gel. Lanes 1–4 contain no competitor; lanes 5–8 contain a 200-fold molar excess of unlabeled B fragment; lanes 9–12 contain a 200-fold molar excess of unlabeled mutant fragment B-C_4A.

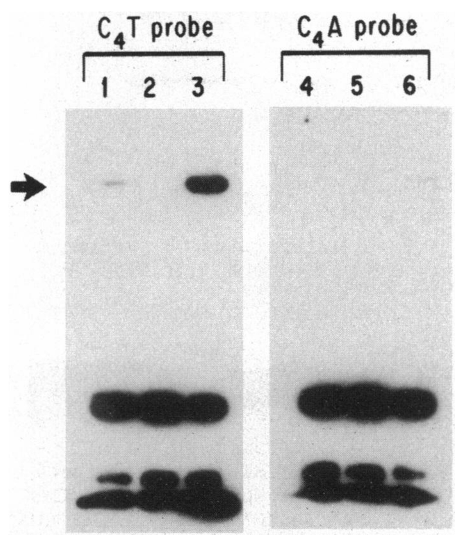

FIG. 3. The levels of a STRE-specific binding protein correlate with expression levels of *MSN2*. Protein extract (60 μg) was fractionated by SDS/PAGE, electrotransferred to nitrocellulose, denatured with guanidine hydrochloride, renatured in binding buffer, and probed with ^{32}P-labeled concatameric DNA. Lanes 1–3, wild-type C_4T probe; lanes 4–6, mutant C_4A probe. Lanes 1 and 4, strain S288C ($MSN2^+$); lanes 2 and 5, strain MCY2144 (*msn2*); lanes 3 and 6, strain MCY2144/pEY32H (*MSN2* overexpression). The arrow marks the position of a ≈97-kDa polypeptide.

protein was absent from extracts prepared from the *msn2* disruption strain (lane 2), although the nonspecific DNA-binding activities remained. Moreover, overexpression of *MSN2* resulted in dramatically higher levels of this C_4T-specific binding activity (lane 3). These result were consistent with the gel mobility-shift data indicating that Msn2p was the principal C_4T-specific binding activity in yeast. Although the calculated molecular weight of Msn2p is ≈78 kDa, this protein's apparent molecular mass was 97 kDa, indicating that it migrates in SDS/PAGE anomolously.

***MSN2* Controls Expression of Yeast Genes in Response to Heat Shock and DNA Damage.** To investigate the role of *MSN2* in the stress regulation of yeast genes, RNA was isolated from cells exposed to either heat shock or DNA damage stress and analyzed by Northern blot hybridization. As shown in Fig. 4, stress-induced transcript levels of three known STRE-regulated genes, *DDR2*, *CTT1*, and *HSP12*, were all greatly reduced in the *msn2* disruption strain relative to wild-type cells. In strain MCY2144/pEY32H, in which *MSN2* is overexpressed from a high copy number vector, there was a significant increase in the basal (uninduced) levels of transcripts from these genes compared with wild-type cells.

Disruption and overexpression of *MSN2* had no effect on stress induction of *SSA3* or *RNR3* transcript levels. *SSA3* encodes a yeast hsp70 protein and requires the heat shock element/heat shock transcription factor pathway for transcription activation in response to heat shock (ref. 16; J. Simon and K.M., unpublished data). The *RNR3* gene encodes a DNA damage-inducible alternate regulatory subunit for the ribonucleotide diphosphate reductase of yeast (17), and, unlike *DDR2*, requires the DUN1 and SAD1/RAD53 kinases for activation (18, 19). As shown in Fig. 4, *SSA3* and *RNR3* transcripts accumulated normally on disruption and overexpression of *MSN2*, confirming the independence of these stress regulatory pathways and demonstrating that altering Msn2p levels in the cell specifically affected expression of STRE-regulated genes.

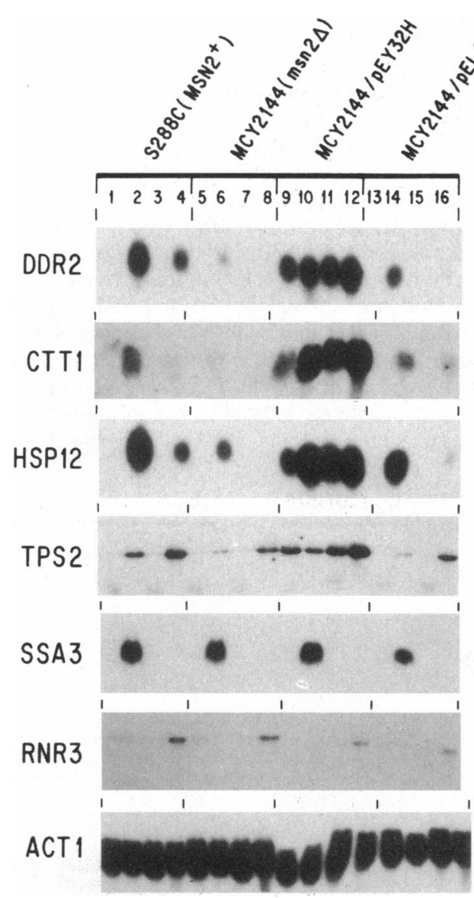

FIG. 4. Stress induction of STRE-regulated genes *in vivo* is controlled by *MSN2*. RNAs were isolated from strain S288C ($MSN2^+$); strain MCY2144 (*msn2*); strain MCY2144/pEY32H (*MSN2* overexpression); and strain MCY2144/pEL45 (*MSN4* overexpression). Cells were grown at 23°C for heat shock controls (lanes 1, 5, 9, and 13); grown at 23°C and stressed with a heat shock for 20 min at 37°C (lanes 2, 6, 10, and 14); grown at 30°C for DNA damage controls (lanes 3, 7, 11, and 15), or grown at 30°C and treated for 1 h with 0.1% methyl methanesulfonate (lanes 4, 8, 12, and 16). RNA samples were electrophoresed and analyzed by Northern blot hybridization using ^{32}P-labeled DNA probes for the indicated genes. The *SSA3* and *RNR3* genes are not regulated through STREs and are shown as controls. The RNAs were hybridized with an *ACT1* probe to verify equal loading.

The results from gel mobility-shift experiments suggested that the *MSN2* homolog, *MSN4*, can weakly interact with the STRE promoter element. Levels of *DDR2*, *CTT1*, and *HSP12* transcripts were examined in *msn2* disruption strain MCY2144/pEL45 containing multiple copies of the *MSN4* gene. In this strain, *MSN4* overexpression partially restored transcriptional activation of the STRE-regulated genes by stress. Although the extent of Msn4p overproduction has not been measured directly, both *in vitro* and *in vivo* evidence suggest that Msn4p could partially compensate for lack of Msn2p.

The *TPS2* gene, encoding trehalose phosphate phosphatase, is very likely under STRE control. This gene is inducible by heat shock and osmotic stress, its induction is influenced by cAMP levels, it has four C_4T elements within its promoter, and a reporter gene driven by a ≈300 bp C_4T-containing segment of the *TPS2* promoter is inducible by both heat shock and osmotic stress (4, 20). The data shown for *TPS2* in Fig. 4 indicate that (*i*) *TPS2* transcripts accumulated following exposure to a DNA damaging agent; (*ii*) the *msn2* mutation significantly reduced stress induction of *TPS2*; and (*iii*) the basal level of *TPS2* RNA was significantly elevated in the *MSN2* overexpressing strain. These results confirm that *TPS2*, like *DDR2*, *CTT1*, and *HSP12*, is regulated through the *MSN2*/STRE pathway. Overexpression of *MSN4* modestly enhanced methyl methanesulfonate induction of *TPS2* transcripts in the *msn2* strain, but did not appear to increase heat shock induction.

***MSN2* Is Required for Most Heat Stress Activation of a C_4T-Driven Reporter Gene.** To quantitatively assess the role of Msn2p for STRE-driven reporter expression after stress, plasmid pCT oligo 31/32 containing a STRE-driven *lacZ* reporter (1) was transformed into wild-type strain M12B and the *msn2* disruption mutant MCY2144 and the levels of β-galactosidase activity were measured following heat shock (Fig. 5). As

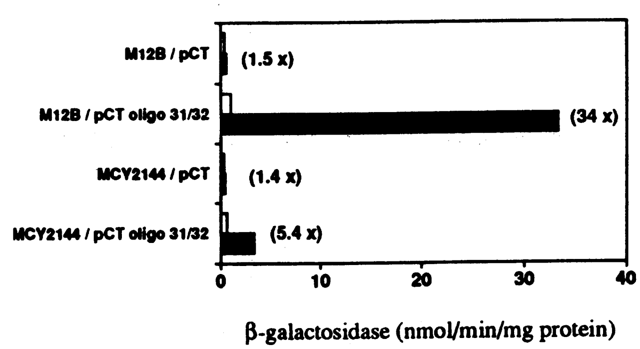

FIG. 5. Most heat shock induction of a STRE-driven *lacZ* reporter gene requires *MSN2*. Strains M12B ($MSN2^+$) and MCY2144 (*msn2*) were each transformed with either plasmid pCT oligo 31/32, which contains *lacZ* driven by the C_4T-containing oligonucleotide 31/32, or plasmid pCT, which lacks the oligo 31/32 insert. Cells were grown at 23°C and either heat shocked for 1 h at 37°C (■) or kept at 23°C (□). β-galactosidase activity was calculated from three independent experiments.

expected, the STRE-containing oligonucleotide 31/32 conferred a significant amount of heat shock inducibility to the reporter gene in *MSN2*$^+$ cells (34-fold increase). In *msn2* disrupted cells, however, this level of induction was reduced by more than a factor of 6, to approximately 5-fold. Thus, Msn2p was necessary for most of the heat shock response mediated by STREs. The lack of any significant induction ($\approx$1.5 fold) in the absence of oligo 31/32 insert argues that the residual 5-fold inducibility observed in the *msn2* mutant was likely due to the action of other transcriptional activators acting through STREs.

DISCUSSION

A *S. cerevisiae* λ gt11 expression library was screened to isolate genes encoding STRE-binding proteins. Of 18 independent positive clones, 14 were derived from the yeast *MSN2* gene and encoded a C-terminal region containing two tandem C_2H_2 zinc-finger DNA-binding motifs. The polypeptide encoded by one of these phages (S2b) specifically bound C_4T-containing DNA during plaque purification (Fig. 1), and *E. coli* lysogenized with this phage expressed a C_4T-binding activity detectable by both Southwestern blot and gel mobility-shift assays (data not shown). *In vitro* binding assays (Figs. 2 and 3) indicated that Msn2p was identical to the previously identified C_4T-binding activity in yeast (2, 3). Extracts prepared from an *msn2* disruption mutant lacked this major C_4T-specific binding activity while those prepared from a strain overexpressing *MSN2* contained a much higher level of this activity. These experimental results argue that Msn2p is the major C_4T-binding activity in yeast.

The role of *MSN2* in the multistress response was directly investigated by analyzing stress induction of yeast genes in strains both lacking and overexpressing *MSN2* (Fig. 4). For all four STRE-regulated genes tested (*DDR2*, *CTT1*, *HSP12*, and *TPS2*) disruption of *MSN2* abolished most heat shock and DNA damage inducibility, whereas *MSN2* overexpression led to nearly constitutive transcriptional activation, in contrast to control cells that had barely detectable basal RNA levels for these genes. *LacZ* reporter data (Fig. 5) concurred with these results in showing that STRE-driven transcription in response to heat shock was greatly curtailed in the absence of *MSN2*. Based on these results, we conclude that Msn2p directly controls heat shock and DNA damage-induced expression of STRE-regulated genes. This conclusion is consistent with previous data showing Msn2p to have a functional transcription activation domain (6).

The mechanism by which Msn2p activates transcription in response to cellular stresses is currently under investigation. We have found no difference in the level of STRE binding in yeast extracts made from stressed versus unstressed cells (ref. 2; unpublished data). Consistent with this observation is the result that Msn2p overexpression activated STRE-regulated genes in the absence of stress. This behavior is similar to that of the yeast *HSF1* and *STE12* genes whose products bind their recognition sequences even when unactivated (21, 22) and give rise to constitutive expression of target genes when overexpressed (23, 24). Heat shock transcription factor is rapidly phosphorylated after heat shock (23), and STE12 protein is phosphorylated after pheromone treatment (22). It has been proposed that phosphorylation might modulate transcription activation directed by these proteins (22, 23, 25). By analogy, Msn2p might undergo a phosphorylation change after stress. Although direct evidence for this hypothesis is lacking, STRE-mediated gene expression can be modulated by mutations affecting the cAMP-dependent protein kinase cascade (3, 26), and osmotic shock induction through the STRE is dependent on the *HOG1* gene, encoding a mitogen-activated protein kinase homolog (27).

Using *lacZ* reporter data, we have determined that nearly 85% of STRE-mediated heat shock induction was *MSN2*-dependent. The *MSN2*-independent induction was likely directed by one or more alternative transcription factors. A candidate for such a factor is the Msn2p structural homolog, Msn4p. We observed that when the *MSN4* gene was present on a high copy plasmid, stress inducibility of STRE-regulated genes in an *msn2* mutant was partially restored (Fig. 4), and levels of STRE-binding activity detected in crude extracts were increased relative to *msn2* cells without plasmid (Fig. 2). Nevertheless, although these data suggest a role for *MSN4* in stress gene control, they are limited to the nonphysiological condition of *MSN4* overexpression. Recent evidence indicates that there is residual stress gene transcript induction in strains disrupted for both *MSN2* and *MSN4*, and extracts prepared from the *msn2 msn4* double mutant contain STRE-specific binding activity (data not shown).

It has been proposed that Msn2p might recognize promoter sequences of the *SUC2* (invertase) gene that normally are bound by *MIG1* repressor (6). This could account for its ability when overproduced to suppress an *snf1* mutation, because *SUC2* derepression requires *SNF1* kinase (15). There are two known *MIG1* binding sites in the *SUC2* promoter region, one containing a CCCCG motif and the other containing a CCCCC motif (28). These sites are both similar to the STRE recognition sequence, adding support to a model in which Msn2p overproduction leads to occupation of *MIG1* binding sites and increased transcription of *SUC2*.

Our identification of *MSN2* should greatly facilitate the characterization of additional genes that are stress-regulated through the *MSN2*/STRE pathway. Important questions can now be addressed regarding the role of STRE-directed gene expression for survival following cellular stress, the molecular mechanism(s) that activate Msn2p in response to stress, and the contributions of other C_4T binding proteins in modulating this important regulatory system.

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